

We built a reconciliation data platform — then used it to test how AI should review code

A real client needed reconciliation data made legible. We shipped a production platform for the **Indigenomics Institute** — and its own **30 merged pull requests** became the industrial dataset for a pre-registered study of multi-agent AI code review.

Research companion: “Evaluating Decorrelation in LLM Code Review: A Pre-Registered Study”

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RESEARCH CONTRIBUTION

More AI reviewers don't help — **decorrelation does**. Agreement predicts truth only when the reviewers are independent.

89%

3 independent families agree → matches ground truth

54%

one model, three runs → mostly noise

01 THE PROJECT — RAP DATA PORTAL

PROBLEM

Companies publish reconciliation commitments. Legal precedents sit scattered across court records. There is no searchable, verifiable, unified view — so nobody can track progress.

OUR SOLUTION

▪ Searchable RAP commitments ▪ Legal precedent database ▪ AI extraction from PDFs ▪ Human review workflow ▪ AWS serverless hosting
▪ Authentication & roles ▪ Progress tracking ▪ Coverage analytics

WHO USES IT

▪ Indigenomics Institute ▪ Indigenous organizations ▪ Corporate partners ▪ Researchers & legal advisors

Indigenomics · RAP Index

INDIGENOMICS RAP Index Organizations Suppliers Coverage analysis Verification Alignment Extract Notifications

Table Explore

The RAP Index commitments by sector, size & type
Reconciliation commitments across the network, and how they progress over time. Distinct from Spend Coverage, which tracks confirmed dollars.

Key takeaways
Plain-language highlights, generated automatically from the data below. Start here.

- 106 commitments across 100 organizations, averaging 63% progress. 0% already confirmed.
- Procurement is the most common commitment type (36), while Employment lags on delivery at 53% average progress.
- Of 29 claimed outcomes, 0% are supplier confirmed, 29 remain self-reported and unverified.
- Needs attention: 31 are past their target year without confirmation, and 43 due in 2026 are behind pace.

Generated from the data below - reflects the current filter.

At a glance
The headlines totals for the current view: commitments, organizations, average progress, and the confirmed share.

106

commitments

100

organizations

63%

average progress

0%

confirmed

Indigenomics · Legal Cases

Legal Cases Cases Find similar Activation Legal info Monitoring Methodology

Unofficial reproductions of public court decisions not legal advice every claim links to its source

Legal cases — economic justice
Canada's Indigenous economic justice case law, searchable and citation-anchored.
Describe your situation to find similar cases →

View as: Indigenous government Legal advisor Corporate / advisory

Precedents affirming your community's economic rights and self-determination. *The same public record, re-rendered for your context — anyone can switch, noting a vision.

Search citation, case name, or full text...

Tier: auto All themes All courts All outcomes

All nations from yr to yr clear

561 results · showing 1-10 · browse · start case

Legal Cases
▪ 561 economic-justice decisions
▪ citation-anchored, role-based views

RAP Index

▪ 106 commitments · 100 organizations
▪ 63% average progress tracked

ARCHITECTURE — SERVERLESS ON AWS

CloudFront

edge →

Next.js

on Lambda →

Textract

OCR →

Bedrock

extraction →

DynamoDB

x6 tables

WHAT THE INSTITUTE CAN NOW DO

▪ See every tracked commitment and its progress in one view
▪ Get a weekly digest of what has gone **overdue**
▪ Trace any claim back to the published source document
▪ Read the legal precedent beside the corporate promise

02 HOW IT BECAME RESEARCH

Client problem

reconciliation data scattered & unverifiable

We built the portal

production platform, shipped on AWS

179 pull requests

30 of them ours — real, merged, AI-reviewed

Benchmark dataset

E1 Qodo · E2 SWE-PRBench · E3 our portal — 2,415 reviews

Four AI review architectures

each rung adds exactly one capability

FOUR AI REVIEW ARCHITECTURES — EACH RUNG ADDS ONE CAPABILITY

Held constant: Same PR snapshot · Same model (Claude Haiku 4.5) · Same prompts · Three runs per arm · Pre-registered on OSF

(a) Agentless — Single LLM Baseline + Sampling

Pull Request (Diff) → Single LLM (One Pass) → Review Report

1 LLM Call 1 Message

(b) Generalists-3 — Compute-Matched Control + Specialization

Pull Request (Diff) → Generalist A → Generalist B → Generalist C → Deterministic Merge (No Communication) → Review Report

3 LLM Calls 3 Messages

(c) Hierarchical — Specialized Coordination + Communication (Manager)

Pull Request (Diff) → Coordinator (Deterministic Routing) → Frontend Specialist → Backend Specialist → Database Specialist → Synthesis & Aggregation (Deterministic) → Review Report

3 LLM Calls 8 Messages

(d) Consensus — Peer Collaboration + Communication (Peers)

Pull Request (Diff) → Protocol Scheduler → Frontend Specialist ↔ Backend Specialist ↔ Database Specialist → Exchange Findings (Challenge & Clarify) → Vote & Resolution (Consensus) → Review Report

9 LLM Calls 22 Messages

03 WHAT WE FOUND · E1 · Q000, 99 PRs

Structure is null. No multi-agent arm beats single-pass F1; specialists tie compute-matched sampling (p=0.73); peer debate adds nothing over a manager (p=0.12).

0.6

0.4

0.2

0

0.487

0.357

0.378

0.369

single-pass F1 = 0.487 — no team beats it

Agentless

1 call

Generalists-3

3 calls

Hierarchical

3 calls

Consensus

9 calls

Extra agents buy recall at a price — up to **5x** the cost per confirmed finding.

Agentless

1 call

0.34

Generalists-3

3 calls

0.45

Hierarchical

3 calls

0.50

Consensus

9 calls

1.76

5x more expensive per confirmed finding than single-pass review

Bounded. Recall plateaus at ≈0.83; on our own unlabeled code, LLM judges disagree (κ=0.03).

WHERE THE SIGNAL LIVES

Agreement recovers **universal functional bugs** far better than project-specific conventions — which is why the ceiling exists.

43%

functional-bug recall at 3 independent families agreeing

18%

rule-violation recall at the same depth

Pair decorrelated review with a deterministic linter.

PROJECT IMPACT

- Production platform **delivered** to the Indigenomics Institute
- Reconciliation commitments made **searchable and verifiable**
- Indigenous legal precedent **explorer**, citation-anchored

RESEARCH IMPACT

- First **compute-matched** comparison of review architectures
- Independence beats more agents** — decorrelate your verifiers
- Guidance for anyone deploying AI code review in production

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Pre-registered on OSF · registration 224vj (embargoed) · github.com/SharonHuang77/rap-review-research